

Presentation 2

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Codes:

For importing datasets:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CarDataImport_Oct21st_Nikole.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/AllCallsDataImport_Oct21st_Nikole.ipynb

Objective 3:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/IdentifyingPreTenantCalls_23Oct_Nikole.R

• What did you do?

I analyzed the Pre-Tenant Menu interactions within the CAR dataset to determine how many callers reached this menu and how many ended the call afterward. I filtered and categorized the call sessions by whether they continued to the Tenant Menu or stopped after the Pre-Tenant Menu, and visualized the distribution of the caller behavior.

• How does it help the project?

This analysis helps evaluate whether the Pre-Tenant Menu message is effective in reducing unnecessary tenant calls. If most callers continue despite hearing the message, it suggests the menu adds little value and could be shortened or removed to improve caller flow and reduce wait times.

• Issues faced (if any)

Some callers had multiple PreTenantMenu entries under the same session ID, often only seconds apart. This made it difficult to determine whether they were re-listening to the same message or restarting the call.

• Attempts to resolve issues (if any)

I addressed this by keeping only distinct session IDs and examining timestamp differences to confirm that duplicates reflected repeated interactions rather than unique calls.

• Issues resolved (if any)

I resolved repetition in PreTenantMenu entries by filtering for distinct contact session IDs, ensuring that each call session was counted only once in the analysis.

• Next steps

Verify operational definitions of “abandonment” by checking the call flow or verbiage script. Suggest to the Legal Aid team to remove or modify their PreTenant Menu and monitor the call patterns again to assess behavioral changes.

Presentation 1

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Codes:

Importing datasets:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CarDataImport_Oct21st_Nikole.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/AllCallsDataImport_Oct21st_Nikole.ipynb

My plot:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Activity%20Pattern_Oct14_Nikole.R

• What did you do?

I helped organize and clean the datasets, collaborating with my team to determine the most effective way to align *All Calls* and *CAR* data for comparison. I created an additional plot showing the hourly share of inbound calls to analyze discrepancies between both datasets.

• How does it help the project?

This work enhances data consistency and visualization, allowing the team to identify mismatches between *CAR* and *All Calls* datasets more effectively, particularly around inbound call patterns and time alignment.

• Issues faced (if any)

The main issue was a mismatch in time variables—specifically, *report_time* from *All Calls* was stored in UTC, while *activity_start_timestamp* in *CAR* appeared to be in Central Time. This caused the hourly distribution of calls to appear misaligned between the two datasets. In addition, the *All Calls* dataset initially included both inbound and outbound calls, leading to further discrepancies since calls from the six key phone lines were not isolated.

• Attempts to resolve issues (if any)

I converted *report_time* from UTC to Central Time (America/Chicago) to align both datasets' timestamps. I also refined the filtering process to include only inbound calls from the six phone lines defined in the code, which correspond to those listed in the *AllCallsData_CARData_overlap.txt* clarification file. By restricting the dataset to calls with direction "TERMINATING" and vendor "CallTower," the data now focuses exclusively on relevant inbound calls.

• Issues resolved (if any)

After converting the timestamps and filtering for inbound calls, the hourly distributions between datasets became consistent. This confirmed that the earlier mismatch was caused by time zone differences and inclusion of outbound calls. The corrections successfully aligned the *All Calls* and *CAR* datasets for accurate comparison, and when the Hourly Share of Inbound Calls plot was generated, both datasets were successfully matched and aligned.

• Next steps

Since the *All Calls* and *CAR* datasets now align closely after filtering and time conversion, my next step will be to interpret the call patterns more deeply. For example, identifying peak hours of inbound activity

and exploring whether certain time periods correspond to higher call volumes or performance differences across the two datasets.